

2.2 Supply

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Learning objectives

2.2 Supply	Depth	Diagrams & calculations
The law of supply—relationship between price and quantity supplied • **Assumptions underlying the law of supply (HL only)** • **The law of diminishing marginal returns** • **Increasing marginal costs**	AO2	
Supply curve	AO4	Diagram: upward-sloping supply curve
Relationship between an individual producer's supply and market supply	AO2	

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Learning objectives

2.2 Supply	Depth	Diagrams & calculations
Non-price determinants of supply • Changes in costs of factors of production (FOPs) • Prices of related goods (in the cases of joint and competitive supply) • Indirect taxes and subsidies • Future price expectations • Changes in technology • Number of firms	AO2	
Movements along and shifts of the supply curve	AO2 AO4	Diagram: movements along and shifts of the supply curve

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What is supply?

Supply is concerned with the behaviour of firms (producers) and refers to the **quantities of a good or service that producers are willing and able to supply at different prices in a given time period.**

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Real world example

Video: How the Tesla Model S is Made

What are some factors that could affect Tesla's ability to produce the Model S?

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Individual supply

A firm's supply for a good can be presented by a supply schedule.

Charlotte's Chocolate Factory's supply schedule for chocolate bars

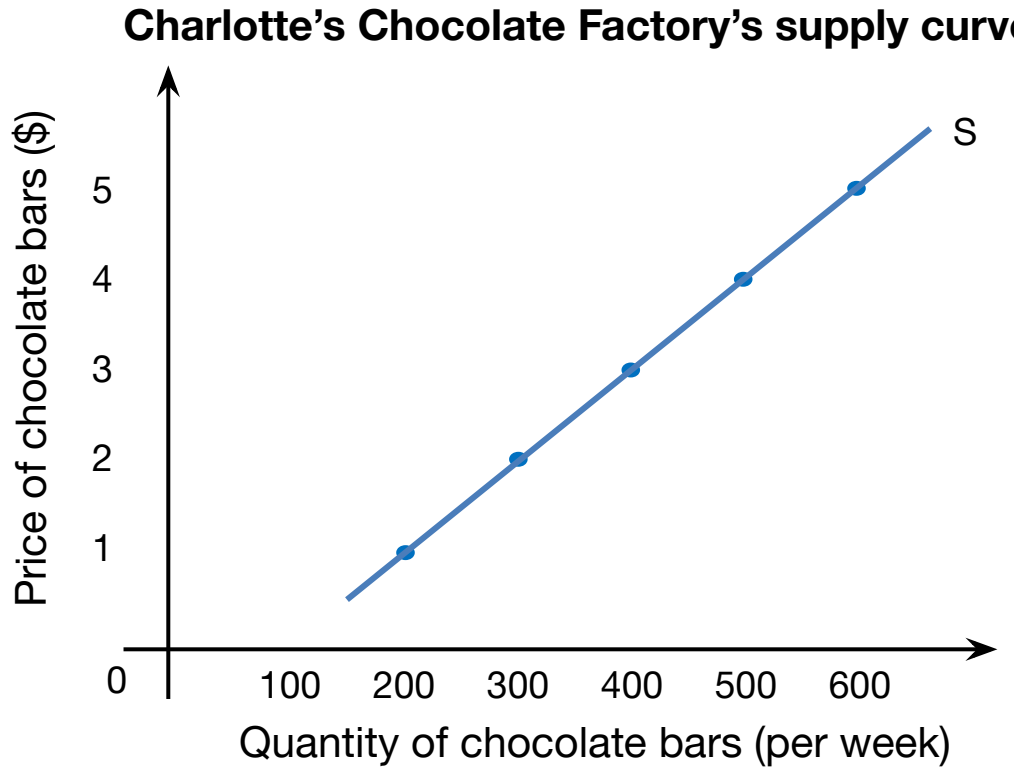
Price of chocolate bars ($)	Quantity of chocolate bars supplied (per week)
$5	600
$4	500
$3	400
$2	300
$1	200

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Individual supply

We can plot the supply schedule on a graph with price on the vertical axis and quantity on the horizontal axis. This is a **supply curve**.

Price of chocolate bars ($)	Quantity of chocolate bars supplied (per week)
$5	600
$4	500
$3	400
$2	300
$1	200



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Individual supply to market supply

Suppose there are four firms in a market for candy bars. The **market supply** is the sum of each individual firm's supply at given prices.

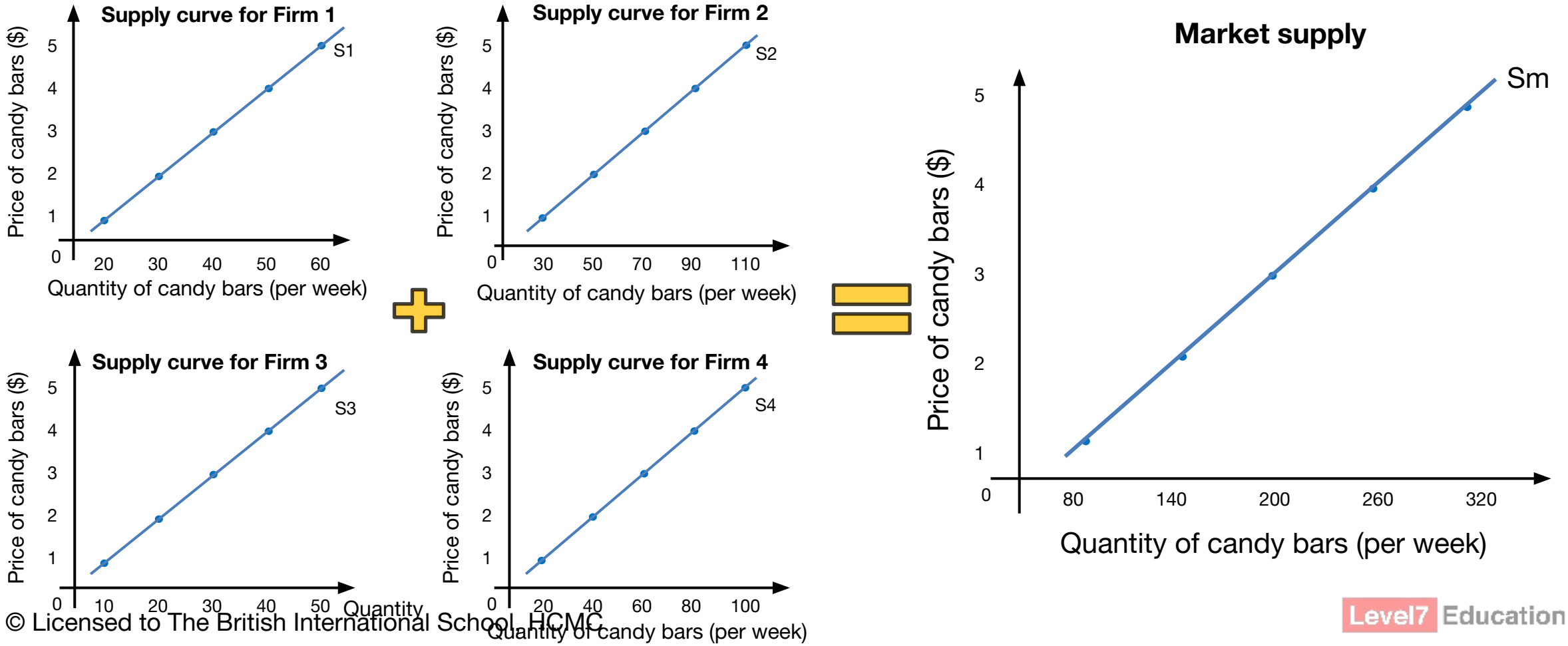
Price of candy bars ($)	Firm 1's Quantity of candy bars supplied	Firm 2's Quantity of candy bars supplied	Firm 3's Quantity of candy bars supplied	Firm 4's Quantity of candy bars supplied	Total quantity of candy bars supplied
$5	60	110	50	100	320
$4	50	90	40	80	260
$3	40	70	30	60	200
$2	30	50	20	40	140
$1	20	30	10	20	80

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Individual supply to market supply

Market supply shows the total quantity of a good or service that firms are willing and able to supply in the market at different possible prices, ceteris paribus.



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Real world example – Research activity

Stockx.com/watches is an online marketplace where buyers and sellers trade watches. We can use real market data to construct a supply curve for any watch.

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Real world example – Research activity

Step 1: Visit stockx.com/watches and select a watch of your choice.

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Real world example – Research activity

Stockx.com/watches

Step 2: For your chosen watch, select "View Asks". This shows the various prices at which potential sellers are willing and able to sell the watch. Using this data, create a supply schedule for the lowest 5 price points. You will need to calculate the cumulative quantity supplied as the price increases. For example:

Bid price	# Available
$1,020	1
$1,040	1
$1,050	1
$1,080	1
$1,090	3

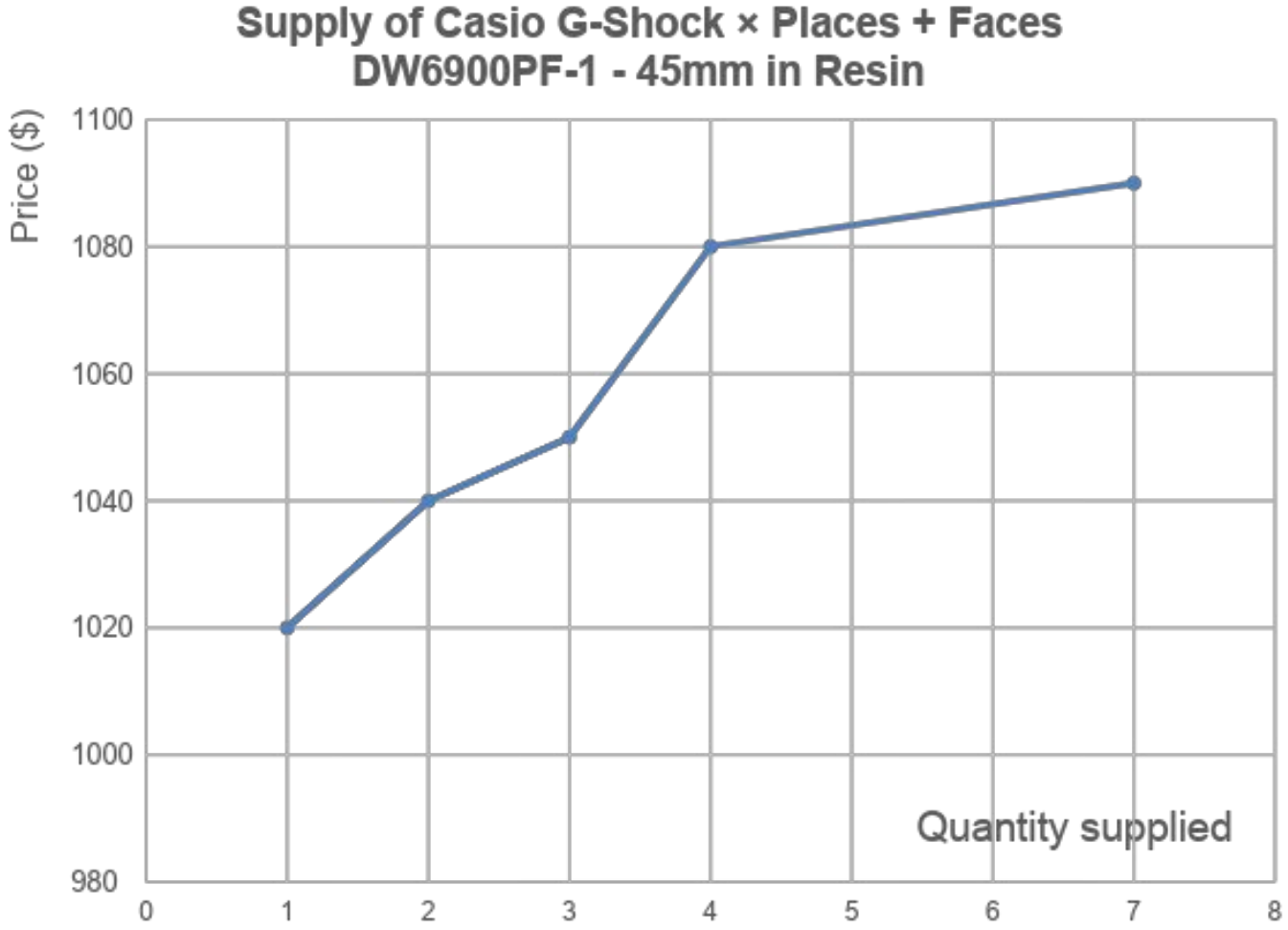
Price ($)	Quantity supplied
1,020	1
1,040	2
1,050	3
1,080	4
1,090	7

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Real world example – Research activity

Step 3: Using your supply schedule, create a supply curve. Label your axis with the appropriate titles and units.

Step 4: Describe the relationship between price and quantity supplied of your chosen watch.

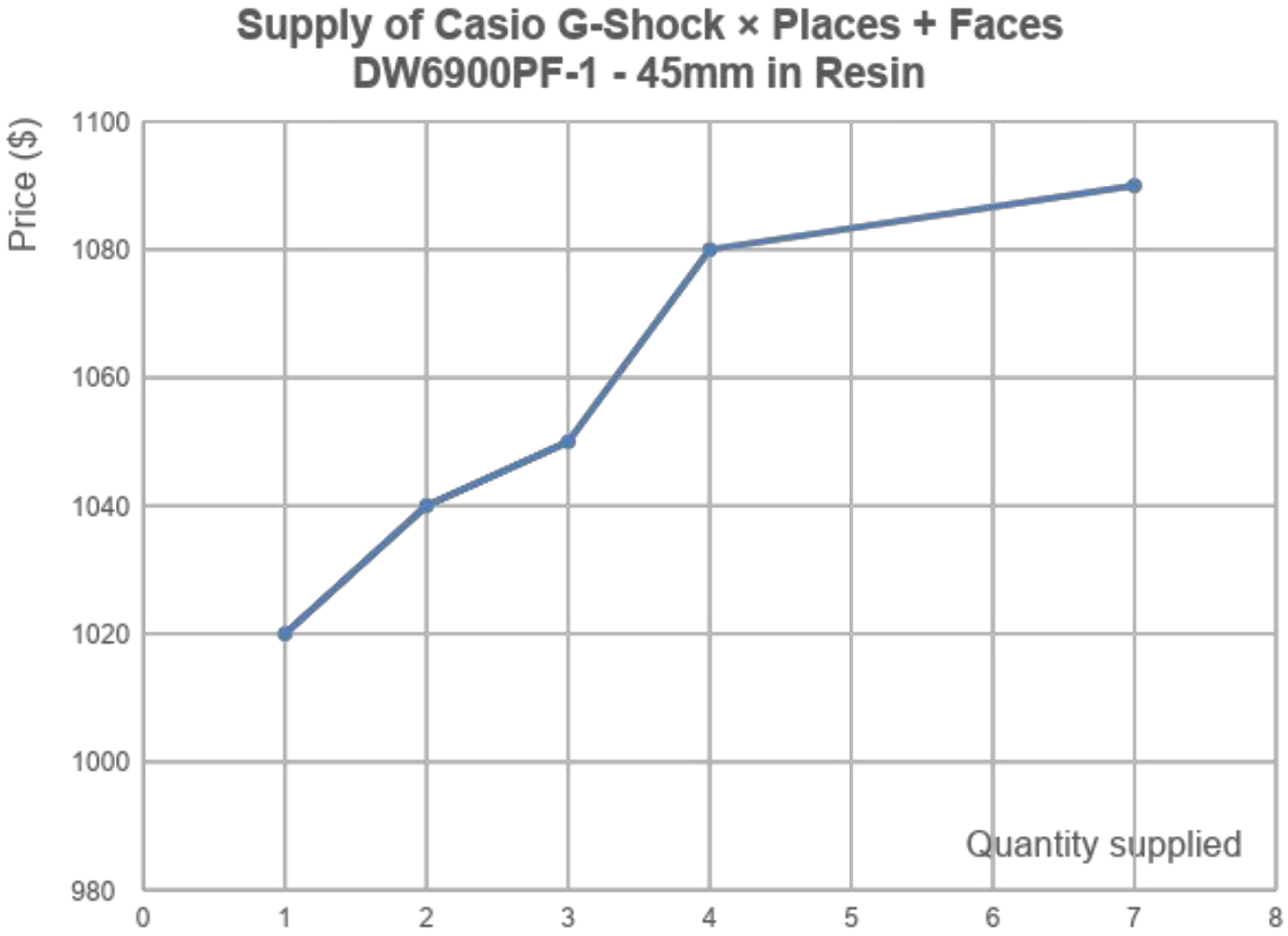


As the price increases, quantity supplied increases.

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The law of supply

The **law of supply** states there is a positive relationship between the price of a good and its quantity supplied over a particular period, ceteris paribus.



What might explain this relationship?

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The law of supply explained

Assuming all else is equal:

Higher prices generally mean that producer profits increase, and so existing firms face an incentive to increase output, while new firms are attracted to enter the market.

Lower prices generally mean lower profitability, so firms have less incentive to produce, thus output decreases.



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Assumptions underlying the law of supply (HL only)

The law of diminishing marginal returns

The law of diminishing returns states that when additional variable factors of production are employed to fixed factors, the marginal returns will eventually decrease. This occurs in the short run, as at least one factors of production is fixed, usually capital.



The **short run** is the period when there is at least one fixed factor of production.

The **long run** is the period when there are no fixed factors of production.

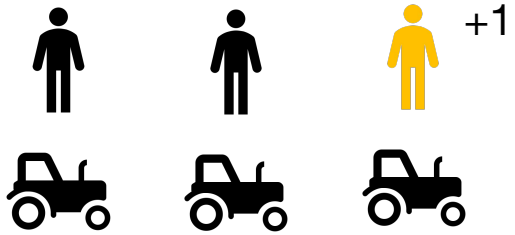
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Assumptions underlying the law of supply (HL only)

The law of diminishing marginal returns - Example

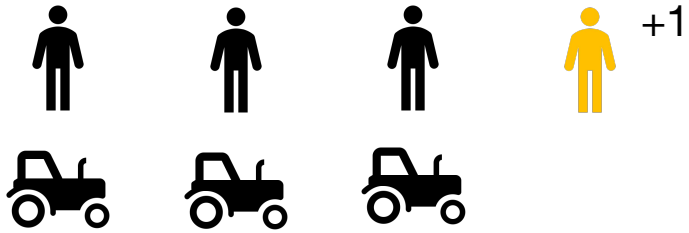
Assume a firm employs workers (variable factors) and machines (fixed factors).

i. Additional workers are employed at first



- Better division of labour
- Machines fully utilized
- Improved efficiency and productivity
- Marginal returns for each additional worker **increases**

ii. More workers employed



- Additional worker has fewer or no fixed factors to use
- Efficiency and productivity of additional worker decreases
- Higher production cost
- Marginal returns for each additional worker **decreases**

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Number of workers (the variable input)	Total product (kilos of potatoes)	Marginal product (kilos of potatoes)
0	0	-
1	20	20
2	50	30
3	90	40
4	120	30
5	140	20
6	150	0

Table 2.4 Total product and marginal product

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Total product (number of units)	Total cost ($)	Marginal cost ($)
1	12	12
2	20	8
3	26	6
4	34	8
5	46	12
6	62	16

Table 2.5: Total cost and marginal cost

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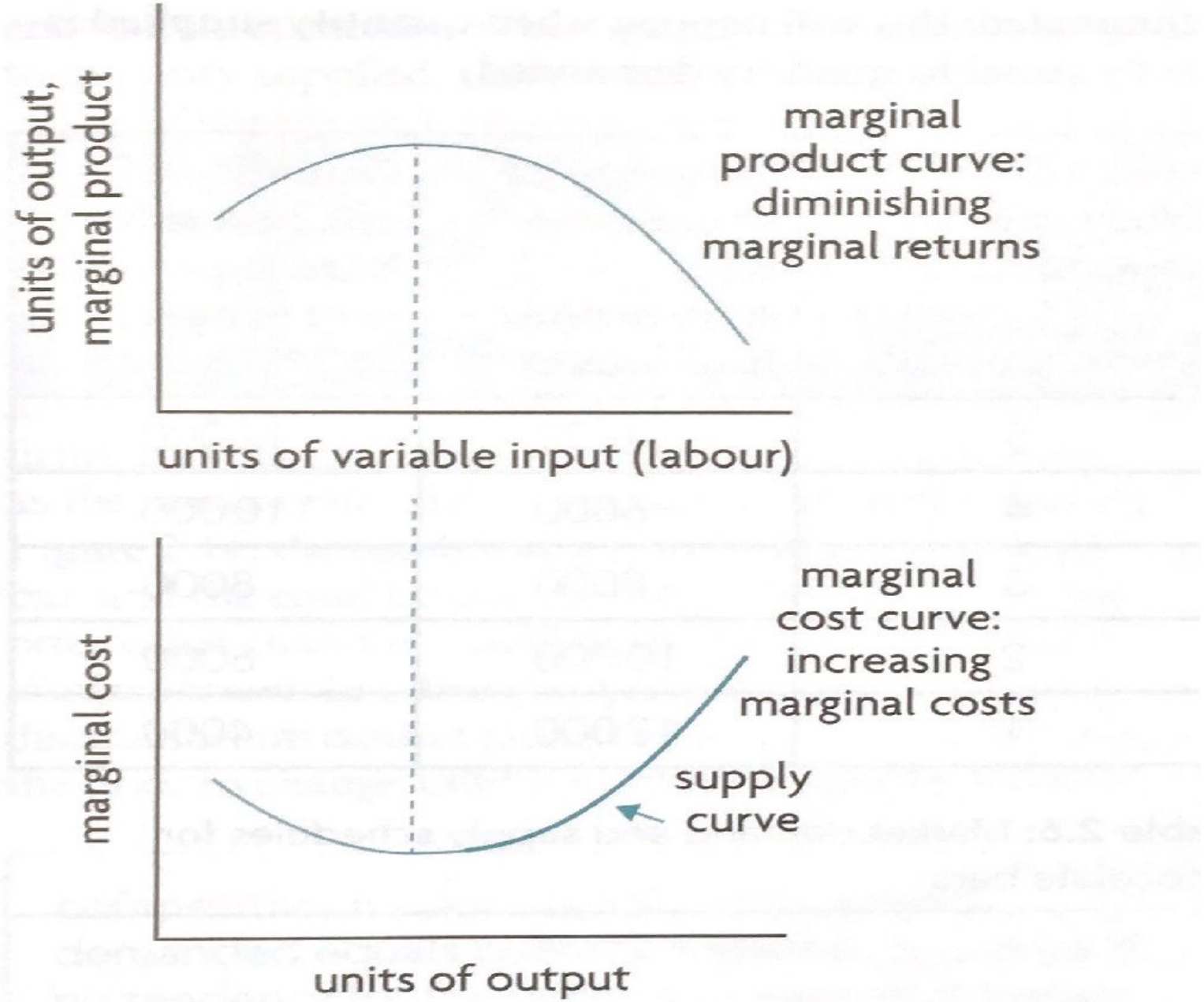


Figure 2.10: Marginal product, marginal cost and the firm's supply curve

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Assumptions underlying the law of supply (HL only)

Increasing marginal costs

Marginal cost is the cost of producing an additional unit of output.

Marginal cost increases as output rises, due to diminishing marginal returns. Firms are only willing to increase output when prices increase in order to cover the higher marginal costs.



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Non-price determinants of supply

Supply shows the quantities of a good or service that producers are willing and able to supply at different prices during a particular time period, **ceteris paribus**.



What other factors are we assuming remains constant?

In other words, what other factors, aside from price, can influence supply?

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Non-price determinants of supply

Non-price determinants of supply are factors that may influence the willingness and ability of producers to supply a good and/or service. This includes but is not limited to:

S - Subsidies and taxes

T - Technology

O - Other related goods' price

R - Resource costs

E - Expectations (of producers)

S - Size of the market



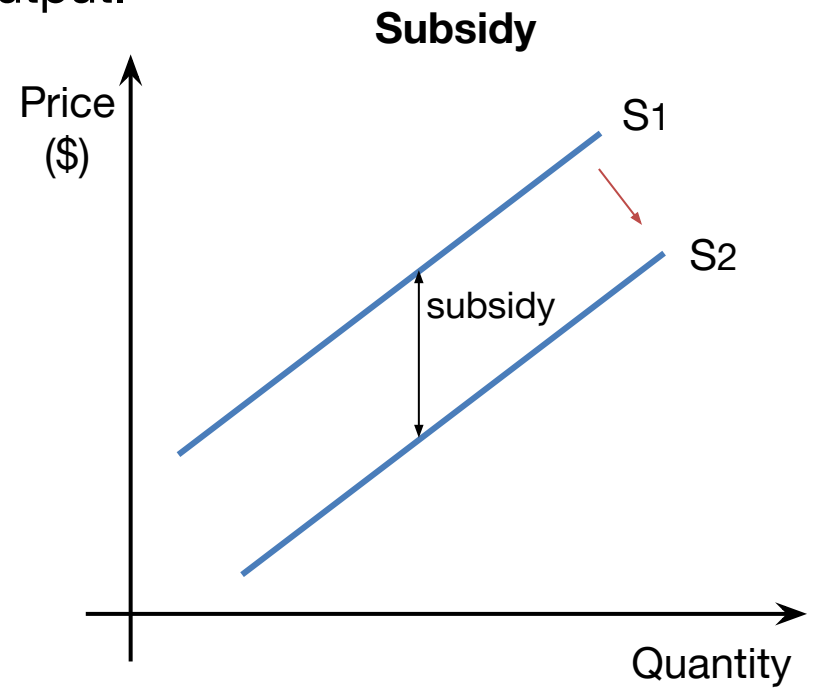
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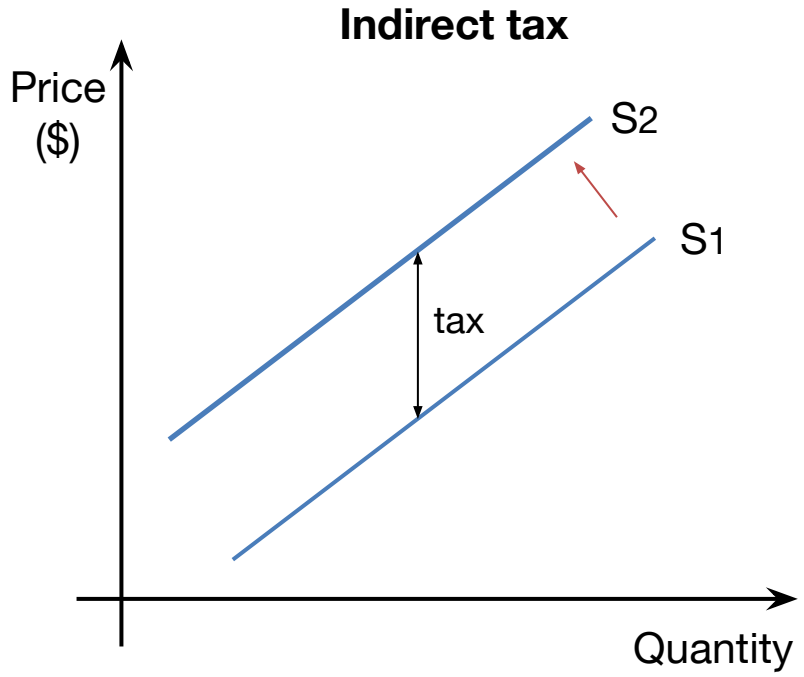
Non-price determinants of supply

Subsidies and indirect taxes

Subsidies are financial aids provided by the government to producers to encourage output.

Indirect taxes are government charges on producers to discourage output.





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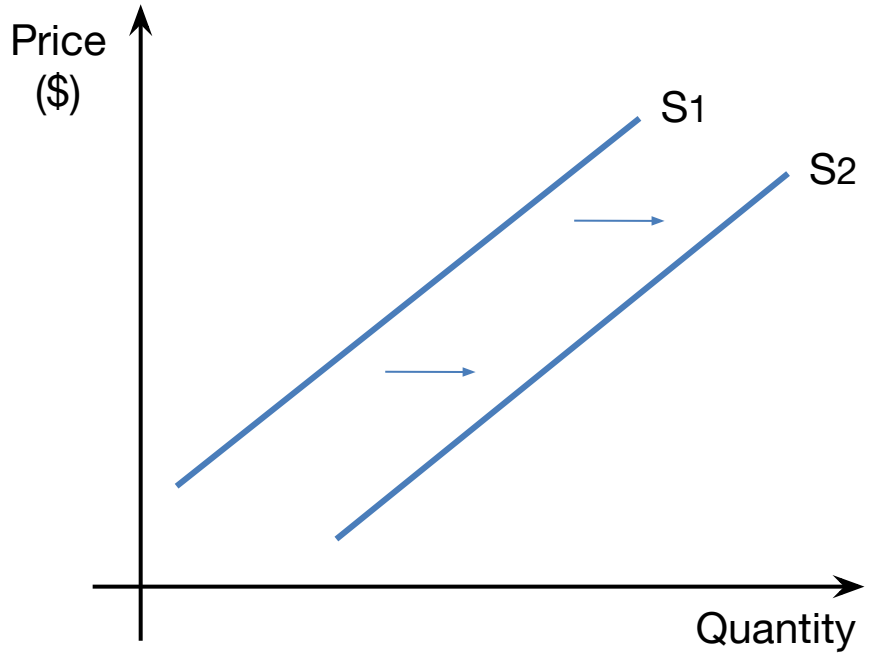
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Non-price determinants of supply

Changes in technology

Advances in technology increase efficiency and productivity, thereby lowering average costs. Supply of goods and services increase at every price level, shifting the supply curve to the right.

For example, e-commerce allows firms to operate at a lower unit cost as hosting a website is cheaper than renting a physical retail outlet.



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Non-price determinants of supply

Competitive supply

A farmer produces both tomatoes and potatoes using shared resources of land and labour.

How would an increase in the price of tomatoes sold impact the supply of potatoes?



Price of tomatoes increases

→ More profitable to produce tomatoes

Farmer allocates more land and labour to harvest tomatoes



Supply of potatoes decreases

→ Supply curve shifts left

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Non-price determinants of supply

Competitive supply

A farmer produces both tomatoes and potatoes using shared resources of land and labour.

How would an increase in the price of tomatoes sold impact the supply of potatoes?



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Non-price determinants of supply

Competitive supply

When two goods are in **competitive supply**, the output of one good is limited by the production of another good due to competing resources. As producers have scarce resources, they cannot produce more of one good without producing less of another.



Brainstorm:

Can you think of other examples of goods in competitive supply?

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Non-price determinants of supply

Joint supply

A farmer produces both lamb and wool from sheep. How would an increase in the price of lamb sold impact the supply of wool?



Price of lamb increases –
Increase in quantity supplied

More sheep for lamb means an increase in the supply of wool



Increase in supply of wool

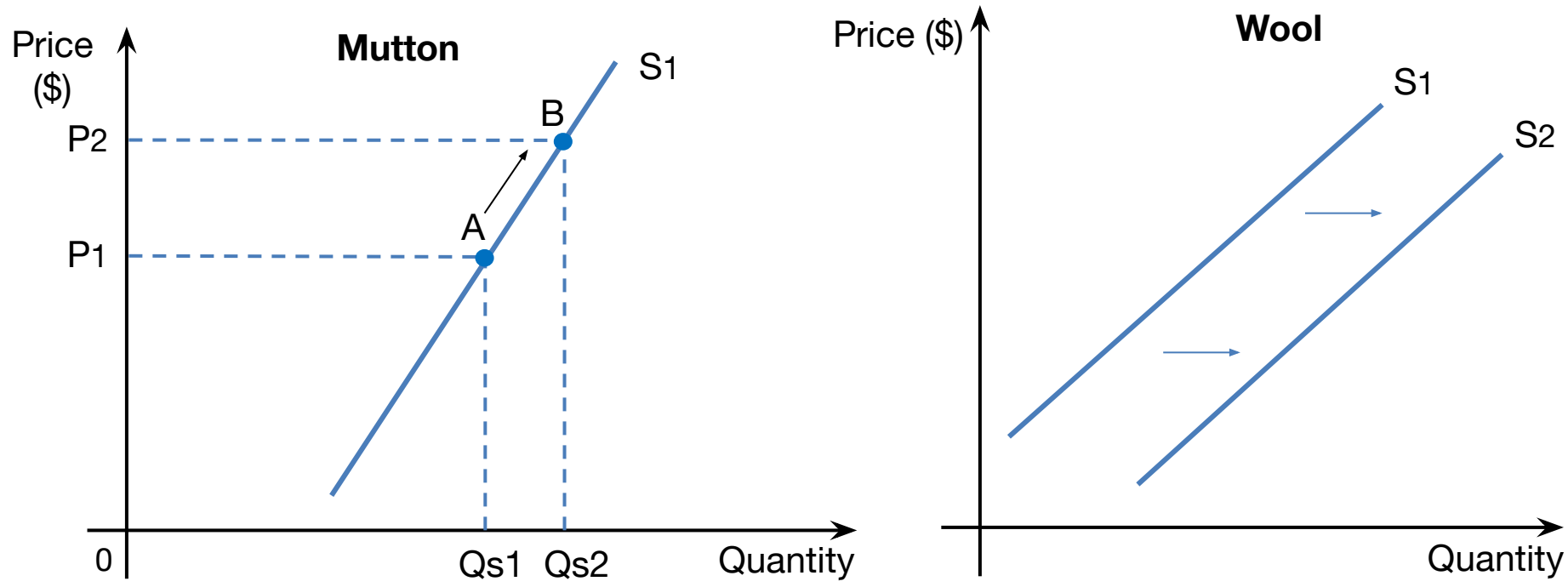
→ Supply curve shifts right

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Non-price determinants of supply

Joint supply

A farmer produces both mutton and wool from sheep. How would an increase in the price of mutton sold impact the supply of wool?



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Non-price determinants of supply

Joint supply

When two goods are in **joint supply**, the increase in production of one good increases the supply of another good. As the supply of the main product changes, the supply of its by-product changes in the same way.



Brainstorm:

Can you think of other examples of goods in joint supply?

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Non-price determinants of supply

Resources costs

Costs of production including wages, rent, interest, and profit, are negatively correlated with supply.

For example, an increase in the minimum wage may reduce the supply of agricultural producers (due to the higher costs of production).



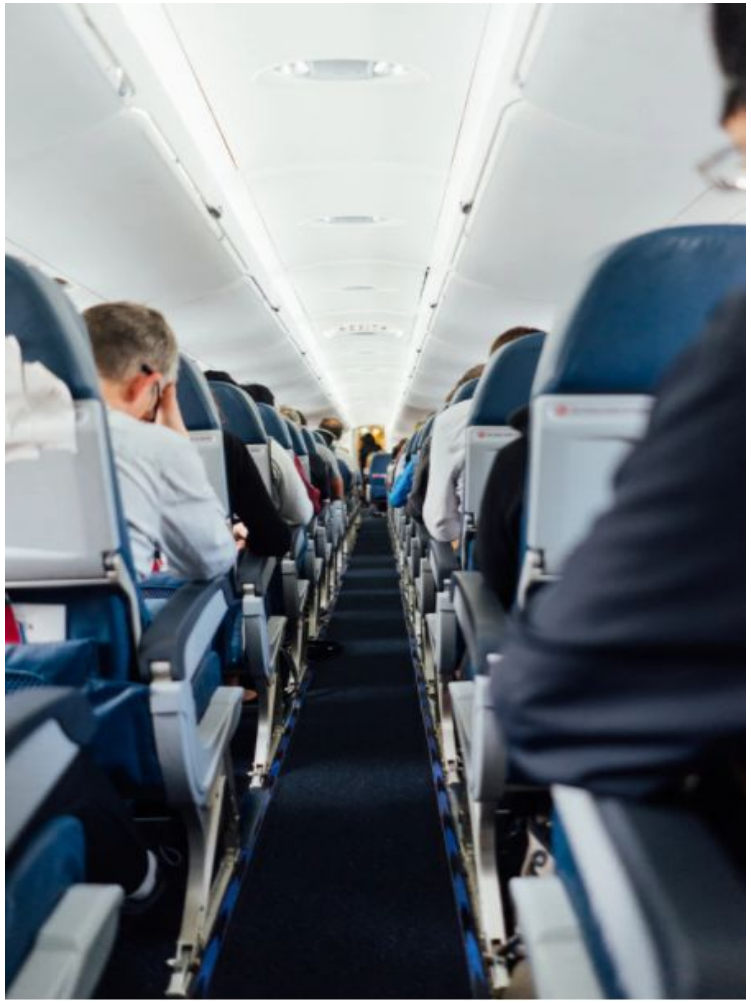
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Non-price determinants of supply

Expectations of producers

Price has a signaling effect in allocating resources to produce goods. There is a positive correlation between future price expectations and the supply of the good.

For example, if an airline company expects demand for flights to worsen during a recession or pandemic, they may decrease the number of flights offered.



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Non-price determinants of supply



Size of market

There is a positive correlation between the number of firms in a market and the market supply of a product.

For example, as there is an increasing number of IB schools around the world, there is a corresponding increase in the market supply of international educational programmes.

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Real World example

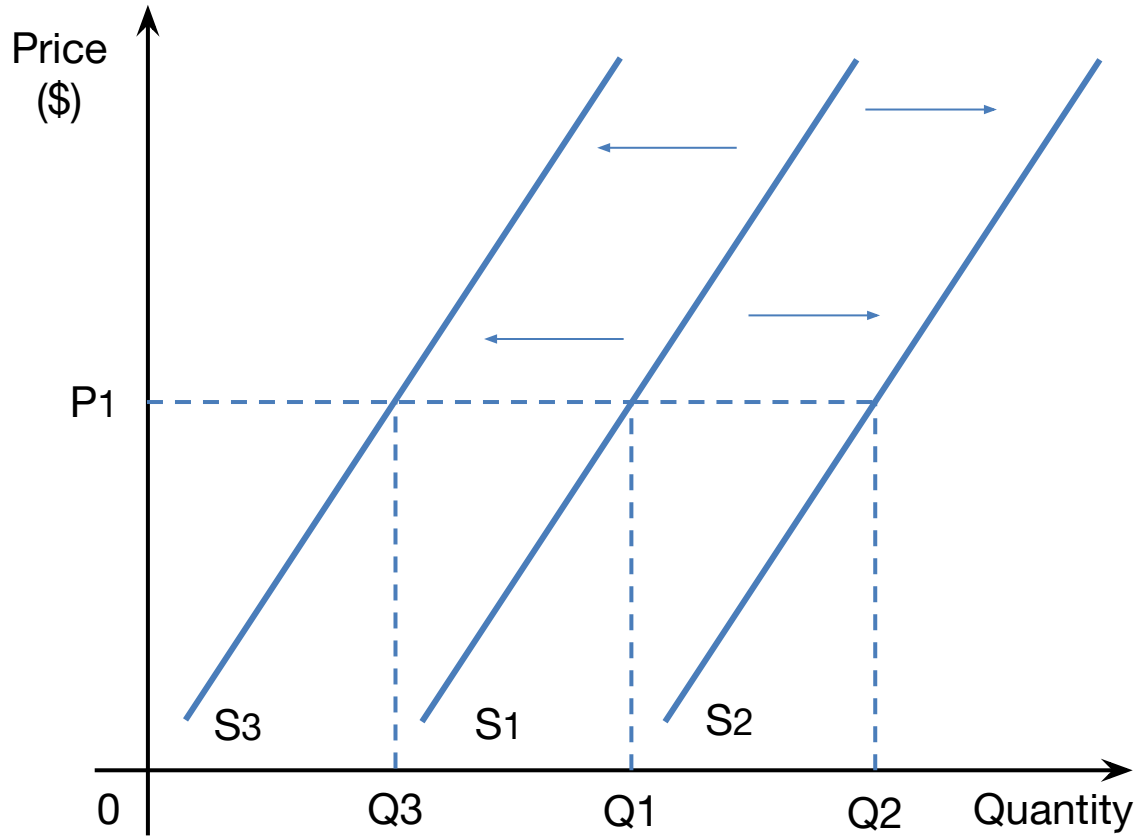
Video: Face mask shortage sparks global race to fulfill orders

How did 3M increase its supply of face masks?

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Shifts in supply

- Changes in non-price determinants of supply cause shifts in the supply curve.
- An outward (rightward) shift of the supply curve (from S1 to S2) indicates that more quantity is supplied at all prices.
- An inward (leftward) shift of the supply curve (from S1 to S3) indicates that less quantity is supplied at all prices.



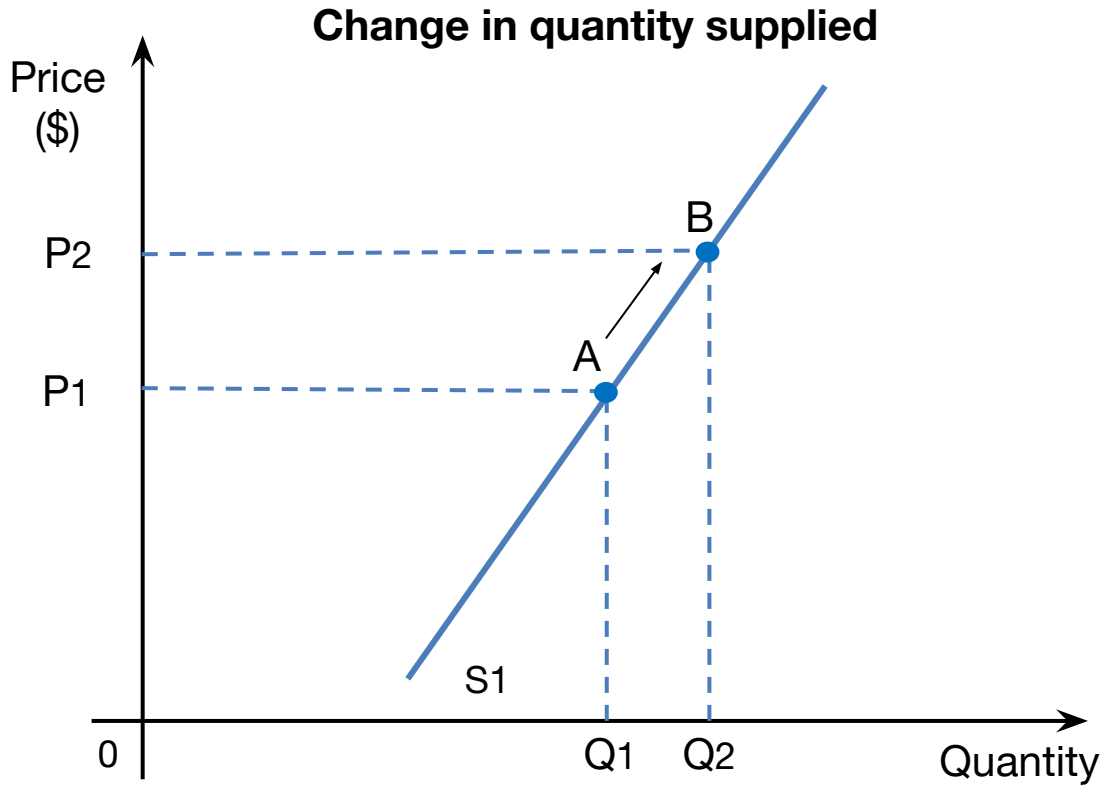
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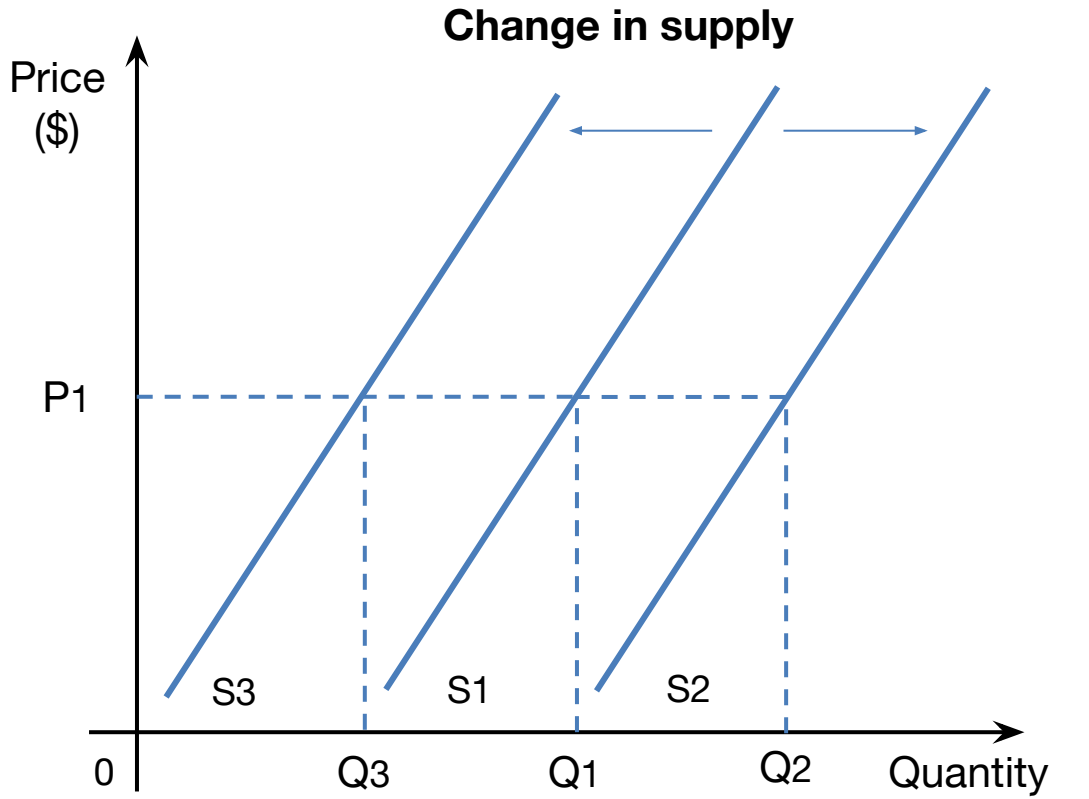
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Movements vs Shifts

Movements along a supply curve are due to price changes, causing *changes in quantity supplied*.

Shifts in a supply curve are due to changes in non-price determinants, causing *changes in supply*.





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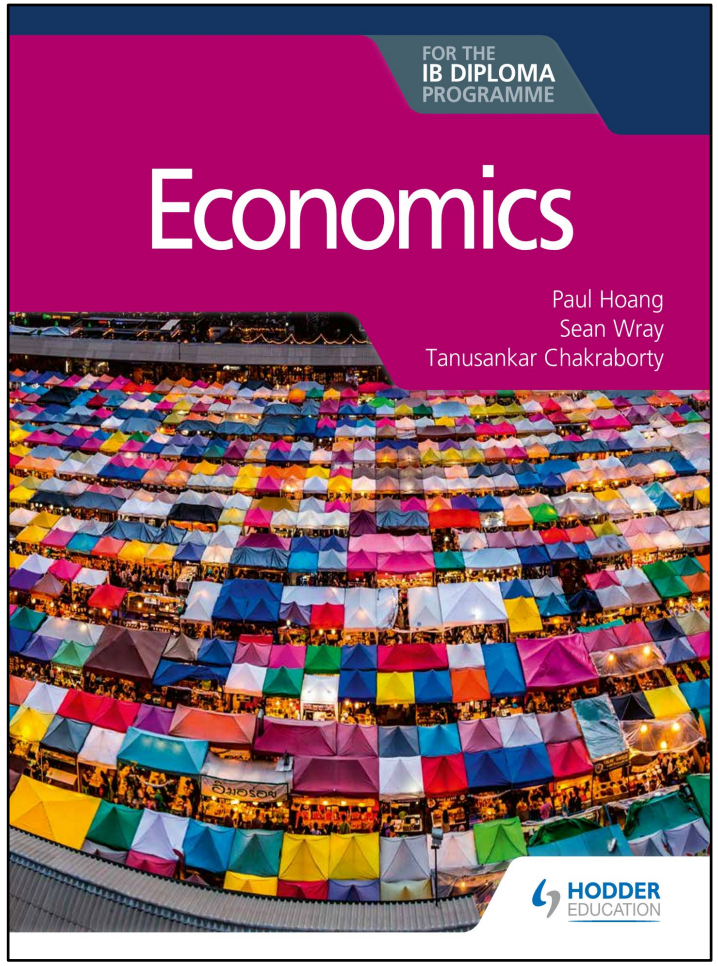
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Over to you…

Hoang, Wray, & Chakraborty (2020)

Economics for the IB Diploma Programme

- Page 65
- Paper 1 Exam Practice Question 4.1
- [10 marks]
- Paper 1 Exam Practice Question 4.2
- [10 marks]



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PAPER 1 EXAM PRACTICE QUESTION 4.1

Explain the difference between a *shift* of the supply curve for a product and a *movement* along the supply of the product. *[10 marks]*

PAPER 1 EXAM PRACTICE QUESTION 4.2

Explain **two** factors which can cause a firm's supply curve to shift to the left. [*10 marks*]

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[Paper 1 Exam practice question 4.1]

Use the assessment criteria on page 1 of the level descriptors. Award up to the maximum marks as indicated. Answers may include:

 Definition(s): Supply and/or supply curve, shift of the supply curve (for a product) and a movement along the supply curve.

 Explanation: of a shift in the supply curve for a good or service being caused by changes in non-price factors that affect the supply of the product, such as taxes, subsidies, adverse weather conditions and production times. Explanation that a movement along the supply curve of a product is caused by changes in prices only – an increase in price causes an expansion in supply, whilst a price reduction causes supply to contract, ceteris paribus.

 Diagram: Supply curve diagram showing *shifts* in the supply curve (due to changes in non-price factors that affect the supply of a product) and *movements* along a supply curve (due to changes in price).

Although the question does not specifically require the use of real-world examples, some students might choose to use appropriate examples to explain their answers. If this is the case, they should be awarded accordingly.

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[Paper 1 Exam practice question 4.2]

Use the assessment criteria on page 1 of the level descriptors. Award up to the maximum marks as indicated. Answers may include:

 Definition(s): Supply and/or supply curve, and shift of the supply curve (for a product).

 Explanation: of any **two** factors that can cause a shift in a firm's supply curve; these must be changes in non-price factors that affect the supply of a product, such as: an increase in indirect taxes, the reduction or removal of subsidies, adverse weather conditions, and prolonged production times (perhaps due to a major natural disaster). The explanation is likely to make reference to and/or consider the ceteris paribus assumption.

 Diagram: Supply curve diagram showing leftward/inwards shifts of the supply curve (due to changes in non-price factors that adversely affect the supply of a good or service).

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External assessment markbands—SL and HL

Paper 1 (SL/HL)

Part (a) 10-mark question

Marks 0–10	Level descriptor
0	The work does not reach a standard described by the descriptors below.
1–2	• The response indicates little understanding of the specific demands of the question. • Economic theory is stated but it is not relevant. • Economic terms are stated but they are not relevant.
3–4	• The response indicates some understanding of the specific demands of the question. • Relevant economic theory is described. • Some relevant economic terms are included.
5–6	• The response indicates understanding of the specific demands of the question, but these demands are only partially addressed. • Relevant economic theory is partly explained. • Some relevant economic terms are used appropriately. • Where appropriate, relevant diagram(s) are included.
7–8	• The specific demands of the question are understood and addressed. • Relevant economic theory is explained. • Relevant economic terms are used mostly appropriately. • Where appropriate, relevant diagram(s) are included and explained.
9–10	• The specific demands of the question are understood and addressed • Relevant economic theory is fully explained. • Relevant economic terms are used appropriately throughout the response. • Where appropriate, relevant diagram(s) are included and fully explained

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Test your knowledge on this unit: Kahoot!

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